

1. Introduction

CLI vs GUI

GUI (Graphical User Interface)

- Uses buttons, icons, menus.
- User interacts using mouse.

CLI (Command Line Interface)

- Uses text commands.
- User interacts using keyboard.

Example

GUI → File Explorer

CLI → Terminal

Terminal Basics

- Terminal is a command line tool.
- Used to run commands.
- Helps interact with computer directly.

Uses

- Create files/folders
- Run programs
- Use Git
- Run Node.js
- Manage projects

Why Terminal is Used

- Faster than GUI
- Important for developers
- Used in servers and deployment
- Needed for Git, Node, React, etc.

Common Developer Tasks

Create project

Run server

Install packages

Push code to GitHub

2. Navigation Commands

pwd

- Shows current folder path.

pwd

Output

/Users/aditya/Desktop

ls

- Shows files and folders.

ls

Output

index.html

style.css

app.js

cd

- Used to change folder.

Move into folder

cd folderName

Move back

cd ..

Go directly to path

cd Desktop/project

3. File & Folder Commands

mkdir

- Creates new folder.

mkdir project

touch

- Creates new file.

touch index.html

rm

- Removes file.

rm index.html

rmdir

- Removes empty folder.

rmdir project

cp

- Copies file/folder.

cp file1.txt file2.txt

mv

- Moves or renames file/folder.

Rename

mv old.txt new.txt

Move file

mv file.txt folder/

4. File Viewing Commands

cat

- Displays file content.

cat index.html

clear

- Clears terminal screen.

clear

5. Paths

Absolute Path

- Complete path from root folder.

/Users/aditya/Desktop/project

Relative Path

- Path relative to current folder.

project/index.html

6. VS Code Terminal

Opening Terminal

Shortcut

Ctrl + `

Menu

Terminal → New Terminal

Running Commands

pwd

ls

node app.js

7. Node Terminal

Running JS Files Using Node

- Node.js runs JavaScript outside browser.

Run File

node app.js

- Executes app.js file.
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Example

app.js

```
console.log("Hello");
```

Terminal

```
node app.js
```

Output

Hello

8. Introduction to Git

Version Control System (VCS)

- Tracks changes in code/files.
- Helps manage project history.
- Allows reverting old changes.

Features

- Save versions
 - Track changes
 - Team collaboration
 - Restore previous code
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Why Git

- Most popular VCS.
- Used by developers worldwide.

Uses

- Track code changes
- Collaborate with team
- Backup code
- Manage different versions
- Connect with GitHub

Benefits

Easy collaboration

Code history

Branching support

Safe backups

9. Git Setup

Install Git

- Download and install Git from official website.

Check Installation

git --version

Configure Git

Set Username

git config --global user.name "Aditya"

Set Email

git config --global user.email "abc@gmail.com"

Check Config

git config --list

10. Repository

Repository

- Folder where Git tracks project files.
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Local Repository

- Repository stored on your computer.

Create Local Repo

git init

Remote Repository

- Repository stored online.
- Usually on GitHub.

Uses

- Backup code
- Share project
- Team collaboration

11. Git Commands

git init

- Initializes Git repository.

git init

git status

- Shows file status.

git status

git add

- Adds files to staging area.

Add Single File

git add app.js

Add All Files

git add .

git commit

- Saves changes permanently.

git commit -m "first commit"

git log

- Shows commit history.

git log

12. Branching

Create Branch

git branch feature

Switch Branch

git checkout feature

Create + Switch

git checkout -b feature

Merge Branch

git merge feature

13. Undo Changes

git restore

- Restores file changes.

git restore app.js

git reset

- Removes staged changes.

git reset

14. .gitignore

- Used to ignore files/folders in Git.
 - Ignored files are not tracked.
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Create .gitignore

touch .gitignore

Ignore Single File

app.js

Ignore Folder

node_modules/

Ignore File Type

*.log

Common Uses

- node_modules
- .env
- Log files
- Temporary files

15. Introduction to GitHub

Git

- Version Control System
- Tracks code changes locally.

GitHub

- Online platform for Git repositories.
- Stores projects on cloud.

Git vs GitHub

Git	GitHub
Tool	Platform
Works locally	Works online
Tracks changes	Hosts repositories

16. GitHub Repository

Create Repository

- Login to GitHub
 - Click New Repository
 - Enter repository name
 - Create repo
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Public Repository

- Anyone can view repository.
-

Private Repository

- Only allowed users can access repository.
-

17. Connecting GitHub with Local Repo

Add Remote Repository

git remote add origin URL

Example

git remote add origin https://github.com/user/project.git

- origin = remote repository name
- Connects local Git repo with GitHub repo

18. GitHub Commands

git push

- Uploads local code to GitHub.

git push origin main

git pull

- Downloads latest changes from GitHub.

git pull origin main

git clone

- Copies repository from GitHub to local machine.

git clone URL

Example

git clone https://github.com/user/project.git

19. Fork & Pull Request

Fork Repository

- Creates personal copy of another repository.
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Create PR (Pull Request)

- Request to merge changes into original repository.

Steps

Fork → Change Code → Push → Create PR

Merge PR

- Accepts and merges pull request changes into repository.

20. GitHub Workflow

Clone → Add → Commit → Push

Clone

- Copy repository from GitHub.

git clone URL

Add

- Add changes to staging area.

git add .

Commit

- Save changes.

git commit -m "message"

Push

- Upload code to GitHub.

git push origin main

21. Deployment Basics

GitHub Pages

- Free hosting service by GitHub.
- Used to deploy static websites.

Static Website Deployment

- Deploy HTML, CSS, JS websites online.

Steps

Push Code → Enable GitHub Pages → Get Live Link